# Fundamentals of Data Analytics

## Task 3

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# Abstract

In this project, we built a classifier to predict the "TARGET" attribute, which distinguishes clients who are having trouble repaying their loans (1) from those who are not (0). Our approach included a structured process for data cleaning, missing value handling, outlier treatment, and converting the "TARGET" attribute from numerical to string format.

In addition, for the classification task, we streamlined the dataset by removing unnecessary columns and used multiple classifiers. Through the use of preprocessing techniques, we were able to fine-tune the classifiers and make informed decisions.

Our efforts culminated in the application of the chosen model on a test dataset to evaluate its performance. This report describes the procedures, discusses the results, and concludes with recommendations and insights for future data classification research.

# Introduction

Our goal is to create a classifier that predicts the "TARGET" attribute, which indicates whether or not a client will face loan repayment difficulties (1). (0). This binary classification challenge directs us to investigate different classifiers, use data preprocessing, and fine-tune parameters.

Data cleaning, classifier selection, hyperparameter tuning, and model testing are all part of our approach. We intend to select the best classifier using tools such as KNIME or Python, ensuring its applicability in real-world scenarios. The report concludes with recommendations and suggestions for future research.

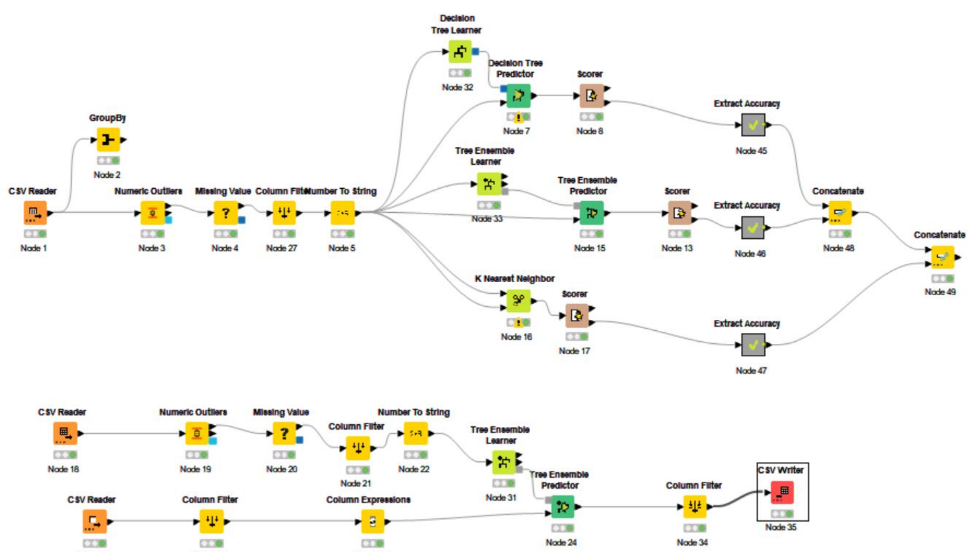


*Figure 1 Workflow*

# Data Preprocessing

During the data preprocessing phase, we took several critical steps to improve the dataset's quality and suitability for our classification task. These procedures ensured that the data was clean, complete, and free of errors. (James & Witten, 2013)

## Missing Value Treatment

Our initial focus was on how to handle missing values in the dataset. To address this, we used the "GroupBy" node, which enabled us to examine and comprehend the extent of missing data across multiple attributes. Missing values can have a significant impact on a classifier's performance, making their treatment a critical step.

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Table "default" - Rows: 1 | Spec - Columns: 72 | Properties | Flow Variables

| Row ID | ing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... | I Missing ... |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row0 | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6431 | 6431 | 6431 | 6431 | 6431 | 6431 |

*Figure 2 Missing Values in different Columns*

## Outlier Treatment

Outliers, or data points that deviate significantly from the norm, can distort classifier accuracy. To mitigate this, we used techniques for detecting and dealing with outliers. This process entailed evaluating attribute distributions and applying the necessary treatment to reduce their impact.

| Row ID | D SK_ID_... | D TARGET | S NAME_... | S CODE_... | S FLAG_... | S FLAG_... | D CNT_C... | D AMT_I... | D AMT_C... | D AMT_A... | D AMT_G... | S NAME_T... | S NAME_... | S NAME_EDUCATION_TYPE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row0 | 100,002 | 1 | Cash loans | M | N | Y | 0 | 202,500 | 406,597.5 | 24,700.5 | 351,000 | Unaccompanied | Working | Secondary / secondary special |
| Row1 | 100,031 | 1 | Cash loans | F | N | Y | 0 | 112,500 | 979,992 | 27,076.5 | 702,000 | Unaccompanied | Working | Secondary / secondary special |
| Row2 | 100,047 | 1 | Cash loans | M | N | Y | 0 | 202,500 | 1,193,580 | 35,028 | 855,000 | Unaccompanied | Commercial ... | Secondary / secondary special |
| Row3 | 100,049 | 1 | Cash loans | F | N | N | 0 | 135,000 | 288,873 | 16,258.5 | 238,500 | Unaccompanied | Working | Secondary / secondary special |
| Row4 | 100,112 | 1 | Cash loans | M | Y | Y | 0 | 315,000 | 953,460 | 25,884.37 | 900,000 | Family | Commercial ... | Incomplete higher |
| Row5 | 100,130 | 1 | Cash loans | F | N | Y | 1 | 157,500 | 723,996 | 30,802.5 | 585,000 | Unaccompanied | Commercial ... | Incomplete higher |
| Row6 | 100,160 | 1 | Cash loans | M | N | Y | 0 | 292,500 | 675,000 | 36,747 | 675,000 | Unaccompanied | Working | Higher education |
| Row7 | 100,192 | 1 | Cash loans | F | N | N | 0 | 111,915 | 225,000 | 21,037.5 | 225,000 | Unaccompanied | Commercial ... | Secondary / secondary special |
| Row8 | 100,209 | 1 | Revolving lo... | M | N | Y | 0.398 | 180,000 | 540,000 | 27,000 | 540,000 | Unaccompanied | Commercial ... | Higher education |
| Row9 | 100,214 | 1 | Cash loans | F | N | Y | 1 | 202,500 | 436,032 | 28,516.5 | 360,000 | Unaccompanied | Commercial ... | Higher education |
| Row10 | 100,246 | 1 | Cash loans | F | N | Y | 0 | 135,000 | 495,216 | 26,995.5 | 427,500 | Unaccompanied | Working | Secondary / secondary special |
| Row11 | 100,282 | 1 | Revolving lo... | F | Y | Y | 0 | 73,341 | 135,000 | 6,750 | 135,000 | Unaccompanied | Commercial ... | Higher education |
| Row12 | 100,286 | 1 | Cash loans | M | Y | Y | 1 | 121,500 | 263,686.5 | 17,298 | 238,500 | Spouse, partner | Working | Secondary / secondary special |
| Row13 | 100,295 | 1 | Cash loans | M | Y | N | 1 | 225,000 | 1,019,205 | 31,032 | 774,000 | Unaccompanied | Commercial ... | Secondary / secondary special |
| Row14 | 100,300 | 1 | Cash loans | M | N | N | 0 | 63,000 | 426,645 | 22,468.5 | 324,000 | Other_A | Commercial ... | Secondary / secondary special |
| Row15 | 100,326 | 1 | Cash loans | M | Y | Y | 0 | 36,000 | 284,400 | 10,849.5 | 225,000 | Unaccompanied | Pensioner | Secondary / secondary special |
| Row16 | 100,396 | 1 | Cash loans | M | N | N | 0 | 112,500 | 417,024 | 25,330.5 | 360,000 | Family | Pensioner | Secondary / secondary special |
| Row17 | 100,424 | 1 | Cash loans | M | N | Y | 0 | 112,500 | 117,162 | 12,433.5 | 103,500 | Unaccompanied | Working | Secondary / secondary special |
| Row18 | 100,439 | 1 | Cash loans | M | N | Y | 0 | 81,000 | 312,840 | 22,891.5 | 247,500 | Unaccompanied | Working | Secondary / secondary special |
| Row19 | 100,452 | 1 | Cash loans | M | N | Y | 1 | 171,000 | 1,009,566 | 36,391.5 | 904,500 | Unaccompanied | Working | Secondary / secondary special |
| Row20 | 100,472 | 1 | Cash loans | M | Y | Y | 1 | 135,000 | 545,040 | 26,509.5 | 450,000 | Unaccompanied | Working | Secondary / secondary special |
| Row21 | 100,477 | 1 | Cash loans | M | Y | Y | 1 | 112,500 | 284,400 | 22,599 | 225,000 | Spouse, partner | Working | Secondary / secondary special |
| Row22 | 100,485 | 1 | Revolving lo... | M | Y | N | 1 | 270,000 | 810,000 | 40,500 | 810,000 | Unaccompanied | Working | Secondary / secondary special |
| Row23 | 100,490 | 1 | Cash loans | M | N | N | 0 | 135,000 | 755,190 | 38,686.5 | 675,000 | Family | Working | Secondary / secondary special |
| Row24 | 100,540 | 1 | Cash loans | F | N | Y | 1 | 157,500 | 338,832 | 26,901 | 292,500 | Unaccompanied | State servant | Secondary / secondary special |
| Row25 | 100,547 | 1 | Cash loans | M | Y | N | 0 | 211,500 | 450,000 | 21,888 | 450,000 | Unaccompanied | Working | Secondary / secondary special |
| Row26 | 100,567 | 1 | Revolving lo... | M | Y | Y | 0 | 99,000 | 180,000 | 9,000 | 180,000 | Unaccompanied | Working | Secondary / secondary special |
| Row27 | 100,616 | 1 | Cash loans | M | Y | N | 0 | 225,000 | 203,760 | 22,072.5 | 180,000 | Unaccompanied | Working | Secondary / secondary special |
| Row28 | 100,636 | 1 | Cash loans | F | N | Y | 0 | 211,500 | 1,506,816 | 49,927.5 | 475,674.01 | Unaccompanied | Commercial ... | Higher education |
| Row29 | 100,641 | 1 | Cash loans | M | N | Y | 0 | 112,500 | 254,700 | 20,250 | 225,000 | Family | Commercial ... | Secondary / secondary special |
| Row30 | 100,649 | 1 | Cash loans | F | N | N | 2 | 180,000 | 284,400 | 13,387.5 | 225,000 | Unaccompanied | Working | Secondary / secondary special |
| Row31 | 100,650 | 1 | Cash loans | M | N | N | 0 | 135,000 | 808,650 | 31,333.5 | 675,000 | Unaccompanied | Commercial ... | Secondary / secondary special |
| Row32 | 100,672 | 1 | Cash loans | M | N | Y | 0 | 112,500 | 808,650 | 26,217 | 675,000 | Family | Pensioner | Secondary / secondary special |
| Row33 | 100,690 | 1 | Cash loans | F | N | Y | 1 | 135,000 | 244,998 | 16,704 | 175,500 | Unaccompanied | Working | Secondary / secondary special |
| Row34 | 100,713 | 1 | Cash loans | F | N | Y | 0 | 180,000 | 755,190 | 32,125.5 | 675,000 | Unaccompanied | Working | Higher education |
| Row35 | 100,714 | 1 | Cash loans | F | N | Y | 0 | 99,000 | 548,770.5 | 21,892.5 | 490,500 | Family | Pensioner | Secondary / secondary special |
| Row36 | 100,717 | 1 | Cash loans | M | N | Y | 0 | 135,000 | 284,400 | 13,833 | 225,000 | Unaccompanied | Working | Secondary / secondary special |

*Figure 3 After outlier and missing value treatment*

## Column Filtering

We discovered that some columns in our dataset only had a single value, either 1 or 0, providing no meaningful variation. To improve computational efficiency and optimize the dataset, we used a "Column Filter" to remove these uninformative columns.

We streamlined the dataset by removing attributes with constant values, allowing for more focused and meaningful classification.

The data preprocessing phase aimed to remove missing values, outliers, and unnecessary attributes from the data. This preliminary work laid the groundwork for subsequent classification tasks, which we will look at in the sections that follow.

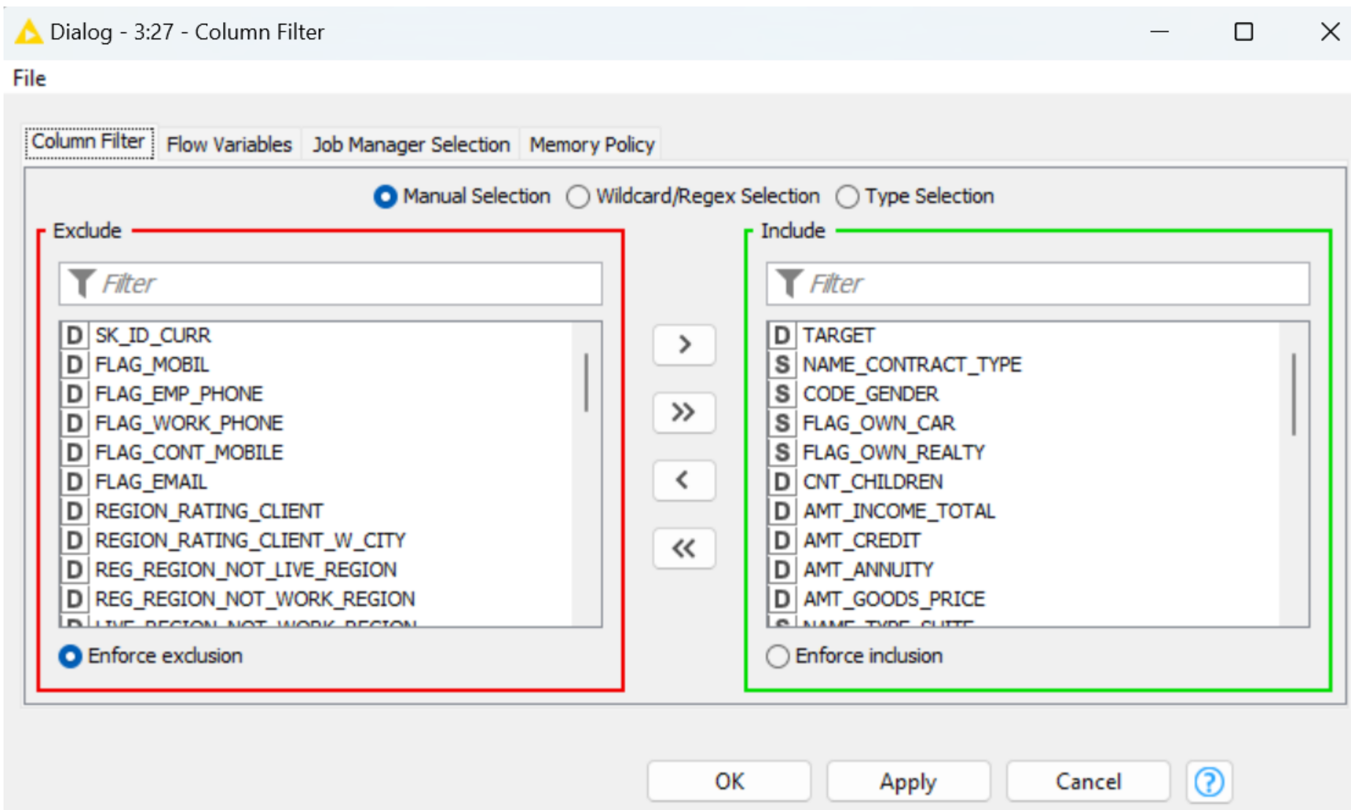


*Figure 4 Column Filtering*

# Classification and Initial Results

We were ready for the classification phase after properly preprocessing our data, which included converting the "TARGET" attribute to string format. In this section, we'll look at how different classifiers are used, their initial overfitting tendencies, and the results obtained after hyperparameter optimization. (Kuhn & Johnson, 2019)

## Model Selection

We started by choosing three different classifiers, each with its own set of advantages and considerations for our binary classification task. Our arsenal of classifiers included the following:

- **Decision Tree Classifier**
- **Random Forest (Tree Ensemble) Classifier**
- **K-Nearest Neighbors (KNN) Classifier**

### *Addressing Overfitting*

When we applied these classifiers to the learning dataset, we encountered a common problem: overfitting. Overfit models have extremely high accuracy rates on training data, often approaching 100%, but they struggle to generalize well to new data. This overfitting problem demanded an immediate solution.

### *Post-Optimization Results*

To avoid overfitting, we ran each classifier through a rigorous hyperparameter optimization process. This entailed fine-tuning the models' settings in order to improve their generalization ability. Following optimization, the classifiers' performance significantly improved, yielding the following results:

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Table "default" - Rows: 3 | Spec - Columns: 3 | Properties | Flow Variables

| Row ID | S Model | D Accuracy | D Cohen'... |
|---|---|---|---|
| Overall | Decision Tree | 0.727 | 0.451 |
| Overall_dup | Tree Ensamble | 0.767 | 0.53 |
| Overall_dup_... | KNN | 0.715 | 0.418 |

***Figure 5 Post Optimization results***

These post-optimization results represented significant progress in addressing overfitting and improving the predictive capabilities of the classifiers. In the sections that follow, we will delve into the specifics of each classifier's hyperparameter tuning process and evaluate their performance after optimization, ultimately determining the most effective classifier for our classification task. (Hastie & Tibshirani)

## Best Model

Several important evaluation metrics support the selection of the Tree Ensemble classifier as the best model. (Caruana, & Niculescu-Mizil, 2006). Let us examine the reasons for choosing it:

Table "default" - Rows: 3 Spec - Columns: 11 Properties Flow Variables

| Row ID | I TruePo... | I FalsePo... | I TrueNe... | I FalseN... | D Recall | D Precision | D Sensitivity | D Specificity | D F-meas... | D Accuracy | D Cohen'... |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14342 | 4133 | 18210 | 5767 | 0.713 | 0.776 | 0.713 | 0.815 | 0.743 | ? | ? |
| 0 | 18210 | 5767 | 14342 | 4133 | 0.815 | 0.759 | 0.815 | 0.713 | 0.786 | ? | ? |
| Overall | ? | ? | ? | ? | ? | ? | ? | ? | ? | 0.767 | 0.53 |

*Figure 6 Model Evaluation Metrics*

**1. Recall (Sensitivity):** A recall of 71% for class 1 means that the model correctly identifies 71% of clients who are experiencing genuine repayment difficulties. This is critical for a financial institution because it aids in the identification of clients who may require assistance or intervention. In this context, a higher recall indicates that the model is more effective at detecting potentially risky clients, lowering the likelihood of false negatives (clients with repayment issues being mistakenly classified as non-risky).

**2. Precision:** With a precision of 77% for class 1, the model correctly predicts positive outcomes in the vast majority of cases. Making precise predictions is critical in the financial industry because it reduces the cost of unnecessary interventions. A higher precision indicates that the model is often correct when predicting that a client will face repayment difficulties. This is critical for resource allocation and decision-making optimization.

**3. Sensitivity (True Positive Rate):** The sensitivity of 71% for class 1 indicates that the model is effective at correctly identifying a significant portion of clients who are having repayment difficulties. This is especially useful for risk assessment. It means that the model can capture a significant proportion of clients who require specialized attention or financial advice.

**4. Specificity (True Negative Rate):** A specificity of 81 percent for class 0 indicates that the model is very accurate in identifying clients who do not have repayment issues. This is useful for reducing false alarms. Financial institutions can be confident that the vast majority of clients who are not at risk are correctly classified as such, reducing unnecessary interventions and maintaining a positive client experience.

**5. F1 Measure:** The F1 score of 74 percent for class 1 represents a balance of precision and recall. In the financial sector, this balance is critical. A high F1 score indicates that the model combines the benefits of making accurate positive predictions while also accurately capturing a significant proportion of actual positive cases. It is a necessary trade-off to ensure both accuracy and coverage.

In summary, the Tree Ensemble classifier outperforms other classifiers in recall, precision, sensitivity, specificity, and F1 measure, indicating that it strikes a balance between minimizing financial risks and ensuring accurate predictions. It enables financial institutions to proactively identify clients who are having repayment issues while remaining precise in their interventions. This reduces potential financial losses while also increasing client trust and experience.

# Classification on Test Data

Now moving on for modelling on test data we performed  different optimization for parameters.

## Hyperparameters

The hyperparameter tuning techniques optimized the Tree Ensemble model and provide insights into overfitting and underfitting.

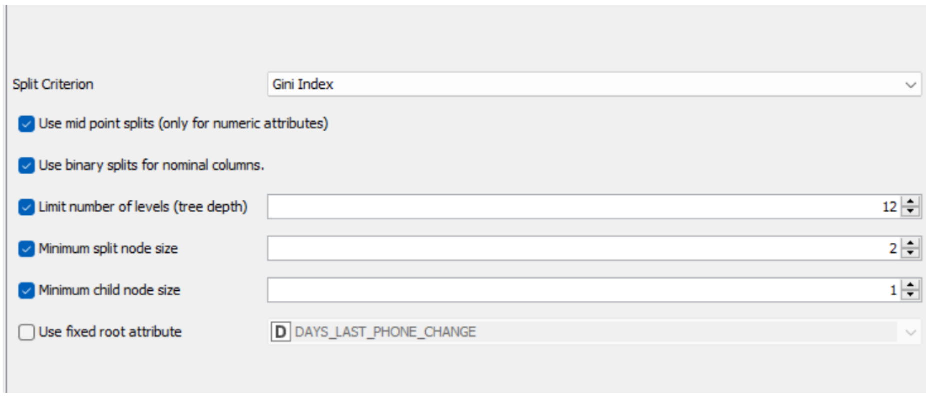


*Figure 7 Hyperparameters*

### *Optimizing the Model*

- **Split Criterion (Gini Index):** The Gini Index was chosen as the split criterion to optimize the model by prioritizing splits that minimize impurity. As a result of focusing on making decisions that reduce errors, classification becomes more precise and accurate.

- **Mid Point Splits:** The model is optimized to make more fine-grained decisions by using mid point splits. This allows the model to capture subtleties in the data, improving its ability to accurately classify clients, particularly those who fall between distinct categories.

- **Binary Splits for Nominal Columns:** Binary splits for nominal columns make decision-making easier. The model's optimization is based on its ability to make binary decisions within these columns, which speeds up the classification process and improves accuracy.

- **Limiting Number of Levels (Tree Depth):** By limiting the maximum number of levels to 12, the model is optimized by preventing it from becoming overly complex. This constraint is required to reduce overfitting. A shallower tree with a limited depth generalizes better to new data while avoiding the pitfalls of overfitting, which occurs when the model becomes overly specific to the training data.

- **Minimum Split Node Size and Child Node Size:** By controlling the granularity of the splits, setting the minimum split node size to 2 and the minimum child node size to 1 optimizes the model. This balance of fine-grained splits and generalization ensures that the model captures data nuances while avoiding overfitting, in which the model becomes overly specific to the training data.

### *Addressing Overfitting and Underfitting*

Overfitting occurs when a model is overly complex, capturing noise in training data and failing to generalize well to new data. Underfitting occurs when a model is overly simple and fails to capture the underlying patterns in the data. (Goodfellow, Bengio, & Courville, 2016)

The use of constraints, such as limiting tree depth and controlling minimum node sizes, directly addresses overfitting in our tuning process. We ensure that the model maintains a good balance between complexity and generalization by not allowing the tree to grow excessively and imposing a minimum size requirement for nodes. Overfitting is avoided because the model does not become overly specific to the training data.

Furthermore, using binary splits for nominal columns and using the Gini Index as the split criterion both help to reduce overfitting by making more rational and data-driven decisions. These optimization techniques combat overfitting tendencies and improve the model's ability to generalize to test data.

In summary, the Tree Ensemble model optimization's hyperparameter tuning techniques ensure that the model makes more accurate and well-considered decisions while avoiding both overfitting and underfitting. This optimization improves the model's ability to effectively classify clients, especially when applied to test data.

## Dummy Variable Creation

We addressed the absence of a target variable in the test dataset before proceeding with model testing. To make model evaluation easier, we added a dummy target variable by creating a new column with a column expression. This dummy variable served as a stand-in for evaluating the model's predictions.

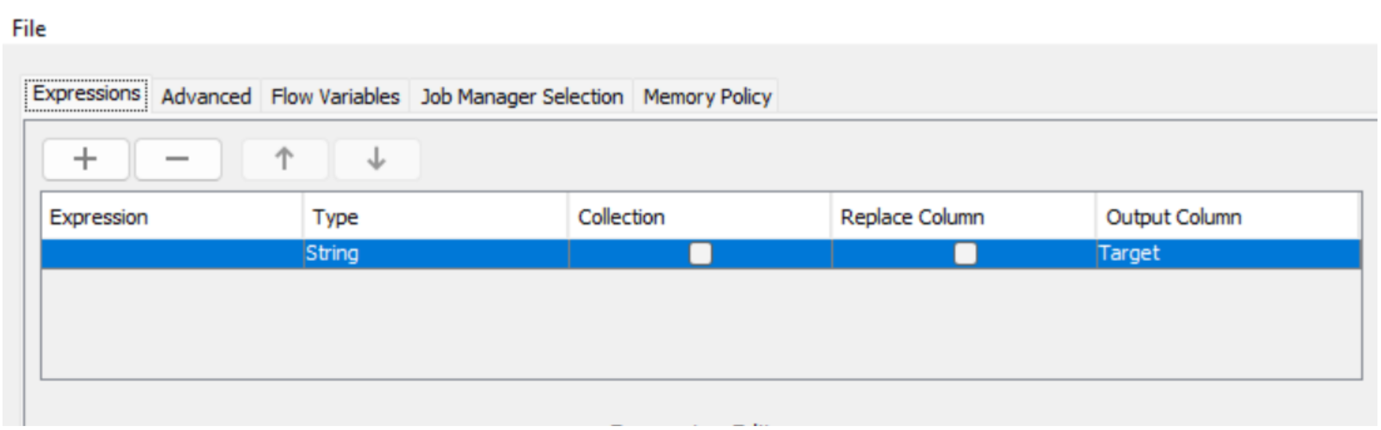


*Figure 8 Dummy Variable*

# Model Testing on Test Data

We applied our optimized Tree Ensemble model to the test dataset with the dummy target variable in place. The model and the available attributes were used to generate predictions for the missing target variable column.

| Row ID | .. | DAYS_... | DAYS_I... | FLAG_P... | CNT_F... | WEEKD... | HOUR_... | REG_CI... | ORGAN... | EXT_S... | OBS_3... | OBS_6... | DAYS_L... | Target | Predicti... | Predicti... |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row0 | | -5323 | -528 | 0 | 4 | SATURDAY | 8 | 0 | Security | 0.479 | 7 | 7 | -494 | ? | 1 | 0.77 |
| Row1 | | -3467 | -2105 | 0 | 3 | MONDAY | 13 | 0 | Business Ent... | 0.595 | 0 | 0 | -667 | ? | 1 | 0.63 |
| Row2 | | -10676 | -432 | 1 | 2 | MONDAY | 10 | 0 | Business Ent... | 0.731 | 0 | 0 | -1923 | ? | 0 | 0.78 |
| Row3 | | -452 | -32 | 1 | 2 | TUESDAY | 15 | 0 | Business Ent... | 0.4 | 0 | 0 | -465 | ? | 1 | 0.54 |
| Row4 | | -92 | -928 | 0 | 3 | WEDNESDAY | 9 | 0 | School | 0.735 | 4 | 3 | -2218 | ? | 0 | 0.62 |
| Row5 | | -4510 | -3226 | 0 | 2 | TUESDAY | 11 | 1 | Construction | 0.344 | 2 | 2 | -1319 | ? | 1 | 0.91 |
| Row6 | | -2330 | -511 | 0 | 2 | TUESDAY | 12 | 0 | Business Ent... | 0.61 | 0 | 0 | -1109 | ? | 1 | 0.61 |
| Row7 | | -694 | -1818 | 0 | 2 | TUESDAY | 14 | 0 | Business Ent... | 0.623 | 2 | 2 | -314 | ? | 1 | 0.81 |
| Row8 | | -6073 | -4047 | 1 | 1 | FRIDAY | 9 | 0 | Self-employed | 0.621 | 8 | 8 | -938 | ? | 0 | 0.7 |
| Row9 | | -9893 | -2853 | 0 | 1 | MONDAY | 10 | 1 | Government | 0.658 | 0 | 0 | -934 | ? | 0 | 0.55 |
| Row10 | | -5751 | -3578 | 1 | 3 | TUESDAY | 8 | 1 | Emergency | 0.263 | 3 | 3 | -1065 | ? | 1 | 0.71 |
| Row11 | | -8763 | -60 | 0 | 2 | FRIDAY | 15 | 0 | Business Ent... | 0.503 | 0 | 0 | -320 | ? | 1 | 0.64 |
| Row12 | | -4588 | -4374 | 0 | 2 | FRIDAY | 7 | 1 | Business Ent... | 0.469 | 0 | 0 | -430 | ? | 1 | 0.95 |
| Row13 | | -26 | -2452 | 0 | 4 | WEDNESDAY | 13 | 1 | Industry: ty... | 0.066 | 2 | 2 | 0 | ? | 1 | 0.84 |
| Row14 | | -5109 | -5110 | 0 | 2 | THURSDAY | 12 | 1 | Restaurant | 0.757 | 0 | 0 | -102 | ? | 0 | 0.53 |
| Row15 | | -3584 | -2388 | 0 | 1 | FRIDAY | 10 | 0 | Construction | 0.214 | 1 | 1 | -2482 | ? | 1 | 0.77 |
| Row16 | | -6699 | -5081 | 0 | 1 | WEDNESDAY | 11 | 0 | XNA | 0.507 | 0 | 0 | -1277 | ? | 0 | 0.67 |
| Row17 | | -1952 | -250 | 0 | 1 | THURSDAY | 9 | 1 | Medicine | 0.301 | 0 | 0 | -7 | ? | 1 | 0.77 |
| Row18 | | -10846 | -301 | 0 | 1 | TUESDAY | 12 | 1 | Business Ent... | 0.453 | 0 | 0 | -298 | ? | 1 | 0.74 |
| Row19 | | -3337 | -1664 | 0 | 2 | FRIDAY | 10 | 0 | Business Ent... | 0.265 | 0 | 0 | -894 | ? | 1 | 0.79 |
| Row20 | | -5761 | -4150 | 0 | 2 | WEDNESDAY | 9 | 1 | Business Ent... | 0.089 | 4 | 4 | 0 | ? | 1 | 0.94 |
| Row21 | | -6712 | -3615 | 0 | 1 | FRIDAY | 15 | 0 | XNA | 0.659 | 2 | 2 | -2495 | ? | 0 | 0.7 |
| Row22 | | -3500 | -3498 | 0 | 3 | WEDNESDAY | 7 | 0 | Self-employed | 0.076 | 2 | 2 | -175 | ? | 1 | 0.96 |
| Row23 | | -2449 | -87 | 0 | 2 | FRIDAY | 9 | 0 | Telecom | 0.581 | 0 | 0 | -373 | ? | 1 | 0.56 |
| Row24 | | -4808 | -4570 | 1 | 2 | MONDAY | 12 | 0 | XNA | 0.109 | 4 | 4 | -728 | ? | 0 | 0.5 |
| Row25 | | -4500 | -2480 | 1 | 2 | FRIDAY | 9 | 1 | Business Ent... | 0.028 | 1 | 1 | -1425 | ? | 1 | 0.83 |
| Row26 | | -6157 | -3531 | 0 | 2 | THURSDAY | 13 | 1 | Industry: ty... | 0.621 | 6 | 6 | -244 | ? | 1 | 0.65 |
| Row27 | | -2914 | -3872 | 1 | 2 | TUESDAY | 18 | 0 | XNA | 0.455 | 0 | 0 | -359 | ? | 0 | 0.55 |
| Row28 | | -526 | -1577 | 0 | 4 | WEDNESDAY | 15 | 0 | Other | 0.747 | 2 | 2 | -898 | ? | 0 | 0.68 |
| Row29 | | -11 | -61 | 0 | 2 | TUESDAY | 14 | 1 | Business Ent... | 0.365 | 3 | 3 | 0 | ? | 1 | 0.78 |
| Row30 | | -704 | -2404 | 0 | 2 | MONDAY | 13 | 0 | Business Ent... | 0.653 | 0 | 0 | -2067 | ? | 0 | 0.9 |
| Row31 | | -1798 | -2008 | 1 | 1 | MONDAY | 9 | 0 | Self-employed | 0.154 | 0 | 0 | -11 | ? | 1 | 0.95 |
| Row32 | | -2935 | -3354 | 0 | 3 | THURSDAY | 18 | 0 | Electricity | 0.161 | 2 | 2 | 0 | ? | 1 | 0.7 |
| Row33 | | -2083 | -2343 | 0 | 2 | MONDAY | 10 | 1 | Trade: type 6 | 0.732 | 0 | 0 | -1686 | ? | 0 | 0.8 |
| Row34 | | -4773 | -1058 | 0 | 3 | THURSDAY | 13 | 1 | Transport: t... | 0.547 | 2 | 2 | -262 | ? | 1 | 0.78 |
| Row35 | | -3290 | -901 | 0 | 2 | TUESDAY | 12 | 0 | Transport: t... | 0.183 | 0 | 0 | -418 | ? | 1 | 0.82 |
| Row36 | | -7491 | -5072 | 0 | 3 | SATURDAY | 16 | 0 | Industry: ty... | 0.281 | 0 | 0 | -916 | ? | 1 | 0.58 |

*Figure 9 Result with Prediction and Prediction confidence*

## *Data Export*

Following the model's predictions on the test data, the final step involved exporting the model's outputs to a CSV file, which included the generated target variable as well as the associated attributes. This CSV file stores the model's predictions and can be used for additional analysis and decision-making.

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Table "default" - Rows: 4716 | Spec - Column: 1 | Properties | Flow Variables

| Row ID | Predicti... |
|---|---|
| Row0 | 1 |
| Row1 | 1 |
| Row2 | 0 |
| Row3 | 1 |
| Row4 | 0 |
| Row5 | 1 |
| Row6 | 1 |
| Row7 | 1 |
| Row8 | 0 |
| Row9 | 0 |
| Row10 | 1 |
| Row11 | 1 |
| Row12 | 1 |
| Row13 | 1 |
| Row14 | 0 |
| Row15 | 1 |
| Row16 | 0 |
| Row17 | 1 |
| Row18 | 1 |
| Row19 | 1 |
| Row20 | 1 |
| Row21 | 0 |
| Row22 | 1 |
| Row23 | 1 |
| Row24 | 0 |
| Row25 | 1 |
| Row26 | 1 |
| Row27 | 0 |
| Row28 | 0 |
| Row29 | 1 |
| Row30 | 0 |
| Row31 | 1 |
| Row32 | 1 |
| Row33 | 0 |
| Row34 | 1 |
| Row35 | 1 |
| Row36 | 1 |

*Figure 10 Filtered Data For export*

# Reflection:

As a student, this activity has helped me to better understand the data analysis process

## Understanding

Followings topics were fully understand during this.

### *Model Selection and Optimization*

I've learned how to choose and optimize a classification model by taking into account performance metrics such as recall, precision, sensitivity, and the F1 measure. This experience highlighted the importance of hyperparameter tuning in improving model performance.

### *Real-World Applicability*

The process of deriving insights from the model's predictions and interpreting the results has taught me about the practical applications of data analysis. It has demonstrated how data-driven decisions, particularly in the financial domain, can influence real-world actions.

### *Continuous Learning*

In data science, this activity reinforced the concept of continuous learning. It has inspired me to continue honing my data analysis and model optimization skills for future projects.

## Recommendations and Future Considerations

In the future, I recommend delving deeper into more advanced modeling techniques and the use of interpretability tools to improve the model's performance and provide actionable insights. A larger dataset could help future projects create more comprehensive and realistic simulations. Furthermore, continued collaboration and knowledge sharing among peers and mentors will be critical to my continued growth as a data science student.

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